

30 Days SQL Micro course Answer PDF

1. Total Number of Patients o Write an SQL query to find the total number of patients across all hospitals.

Answer: -

```
SELECT
    COUNT(*) AS TOTAL_PATIENTS
FROM
    HOSPITAL_DATA;
```

2. Average Number of Doctors per Hospital
o Retrieve the average count of doctors available in each hospital.

Answer

```
SELECT
    HOSPITALNAME,
    AVG(DOCTORSCOUNT)
FROM
    HOSPITAL_DATA
GROUP BY
    HOSPITALNAME;
```

3. Top 3 Departments with the Highest Number of Patients
o Find the top 3 hospital departments that have the highest number of patients.

answer

```
SELECT
    DEPARTMENT,
    SUM(PATIENTSCOUNT) AS TOTAL_PATIENTS
FROM
    HOSPITAL_DATA
GROUP BY
    DEPARTMENT
ORDER BY
```

```
TOTAL_PATIENTS DESC
LIMIT
3;
```

4. Hospital with the Maximum Medical Expenses

* Identify the hospital that recorded the highest medical expenses.

```
Answer SELECT
        HOSPITALNAME,
        MAX(MEDICALEXPENSES) AS
MAXMEDI_EXPENSES
FROM
        HOSPITAL_DATA
GROUP BY
        HOSPITALNAME
ORDER BY
        MAXMEDI_EXPENSES DESC
LIMIT
1;
```

5. Daily Average Medical Expenses

* Calculate the average medical expenses per day for each hospital.

```
Answer SELECT
        ADMISSIONDATE,
        AVG(MEDICALEXPENSES) AS
AVG_DAILY_EXPENSES
FROM
        HOSPITAL_DATA
GROUP BY
```

```
ADMISSIONDATE
ORDER BY
AVG_DAILY_EXPENSES DESC;
```

6. Longest Hospital Stay

* Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

Answer

```
SELECT
    HOSPITALNAME,
    MAX(DISCHARGEDATE - ADMISSIONDATE) AS
LONGEST_STAY
FROM
    HOSPITAL_DATA
GROUP BY
    HOSPITALNAME
ORDER BY
    LONGEST_STAY DESC;
```

7. Total Patients Treated Per City

* Count the total number of patients treated in each city.

Answer

```
SELECT
    LOCATIONS,
    SUM(PATIENTSCOUNT) AS TOTAL_PATIENTS
FROM
    HOSPITAL_DATA
GROUP BY
    LOCATIONS
```

ORDER BY

TOTAL_PATIENTS DESC;

8. Average Length of Stay Per Department

* Calculate the average number of days patients spend in each department.

Answer

```
SELECT
    DEPARTMENT,
    AVG(DOCTORSCOUNT) AS AVERAGE_LENGTH
FROM
    HOSPITAL_DATA
GROUP BY
    DEPARTMENT;
```

9. Identify the Department with the Lowest Number of Patients

* Find the department with the least number of patients.

Answer

```
SELECT
    DEPARTMENT,
    SUM(PATIENTSCOUNT) AS TOTAL_PATIENTS
FROM
    HOSPITAL_DATA
GROUP BY
    DEPARTMENT
ORDER BY
    TOTAL_PATIENTS ASC
LIMIT
    1;
```

10. Monthly Medical Expenses Report

• Group the data by month and calculate the total medical expenses for each month.

Answer

```
SELECT
    TO_CHAR(ADMISSIONDATE, 'YYYY-MM') AS
MONTH_YEAR,
    SUM(MEDICALEXPENSES) AS
TOTAL_MONTHLY_EXPENSES
FROM
    HOSPITAL_DATA
GROUP BY
    MONTH_YEAR
ORDER BY
    MONTH_YEAR;
```